Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K_4 Fe (CN)_6. After filtering, the precipitate is dried and weighed. The reactions which occur are ZnS + 2HCl \rightarrow ZnCl_2 + H_2 S 2ZnCl_2 + K_4 Fe (CN)_6 \rightarrow Zn_2 Fe (CN)_6 + 4 KCl If a 2 gram sample of rock yields 0.969 gram of Zn_2Fe(CN)_6, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn_2 Fe (CN)_6 = 342.6.

Answer Choices:

A. 10 % Zn
B. 22 % Zn
C. 20 % Zn
D. 19 % Zn
E. 15 % Zn
F. 18 % Zn
G. 16 % Zn
H. 14 % Zn
I. 12 % Zn
J. 25 % Zn

Reasoning + Answer:

<tool_call> reasoning process here

We are given:

- A 2 gram sample of rock yields 0.969 grams of ZnâĤĤFe(CN)âĤĨ.
- We are to find the percentage of Zn in the original rock sample.
- The molecular weight of ZnâĤĤFe(CN)âĤĨ is given as 342.6 g/mol.
- The atomic weight of Zn is 65.4 g/mol.

Step 1: Find how many moles of ZnâĤĤFe(CN)âĤĨ were produced.

Moles of ZnâĤĤFe(CN)âĤĨ = mass / molecular weight
= 0.969 g / 342.6 g/mol
âīĪ 0.002828 mol

Step 2: Each mole of ZnâĤĤFe(CN)âĤĨ contains 2 moles of Zn.

So, moles of Zn = 2 ÃĹ 0.002828 = 0.005656 mol

Step 3: Mass of Zn = moles ÃĹ atomic weight
= 0.005656 mol ÃĹ 65.4 g/mol
âīĪ 0.3703 g

Step 4: Percentage of Zn in the original sample = (mass of Zn / mass of sample) ÃĹ 100
= (0.3703 / 2) ÃĹ 100
âīĪ 18.515%

So, the percentage of Zn is approximately 18.5%, which rounds to 18%.

<answer>18% Zn</answer>